Point Processing, Histogram Equalisation and CLAHE

1. Point Processing

Point processing is a fundamental image processing technique where each pixel is transformed independently using a mapping function $s = T(r)$.

- Each pixel is processed individually
- Does not consider neighboring pixels
- Fast and easy to compute
- Examples include brightness adjustment, contrast stretching, and thresholding

2. Histogram Equalisation

Histogram equalisation enhances image contrast by redistributing pixel intensities so that the histogram becomes more uniform.

- Uses histogram (intensity distribution)
- Improves global contrast
- Simple and widely used technique

3. CDF Derivation

The cumulative distribution function (CDF) is used to map original pixel values to new values:

s_k = sum_{j=0}^{k} p(r_j)

- p(r_j) is the probability of intensity level r_j
- CDF accumulates probabilities
- Used as transformation function in histogram equalisation

4. Worked Example

Example histogram probabilities: 0.1, 0.2, 0.4, 0.3

CDF becomes: 0.1, 0.3, 0.7, 1.0

- Step 1: Compute histogram
- Step 2: Normalize to probabilities

- Step 3: Compute CDF
- Step 4: Map values to new intensities

5. CLAHE (Adaptive Improvement)

CLAHE improves histogram equalisation by applying it locally and limiting contrast.

- Divides image into tiles
- Applies histogram equalisation locally
- Uses clip limit to prevent over-enhancement
- Interpolates between tiles

6. Failure Cases

Histogram Equalisation Failures:

- Over-enhancement
- Brightness shift
- Loss of detail
- Poor results with uneven lighting

CLAHE Failures:

- Noise amplification
- Sensitive to parameter tuning
- Boundary artifacts between tiles
- May still over-enhance certain regions